

JC4003 Natural Language Processing

Lecture 1: Introduction

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Outline

- What is natural language
- What is Natural Language Processing (NLP)
- Course arrangements
- Practice arrangements
- Assessment methods

Natural Language

Natural Language

- Natural language is the language used by humans (or animals?) in daily communication, encompassing both **spoken** and **written forms**. It has **complex** grammar, semantics, and pragmatic rules, capable of conveying rich emotions, intentions, and information.
- Natural language has **evolved alongside the evolution of human society**.
- Unlike artificial languages, Natural language structure and expressions have evolved naturally over long periods of human interaction, possessing a high degree of flexibility and diversity.
- Natural language examples: English, Chinese, Oracle, ..., Animal languages?

Artificial Language Examples

Esperanto: Created by L. L. Zamenhof in 1887, it aims to be a global auxiliary language.

Ido: A derivative of Esperanto, designed to improve its grammar and vocabulary.

Klingon: Created by Marc Okrand for the Star Trek series as an alien language.

Sindarin / Quenya: Created by J.R.R. Tolkien for the Elvish languages in The Lord of the Rings series.

Na'vi: Created by Paul Frommer for the film Avatar.

Dothraki / Valyrian: Created by David J. Peterson for the Game of Thrones series.

Lojban: Created by the Logical Language Group in 1987 to be a language with complete unambiguity.

Toki Pona: Created by Sonja Lang in 2001, it aims to express basic concepts through minimalism.

Natural Language vs. Artificial Language

Natural Language

- Natural languages arise naturally with the evolution of human society.
- high degree of flexibility, diversity, and for ambiguity and vagueness
- Grammar rules are usually not strict, with exceptions and irregularities.

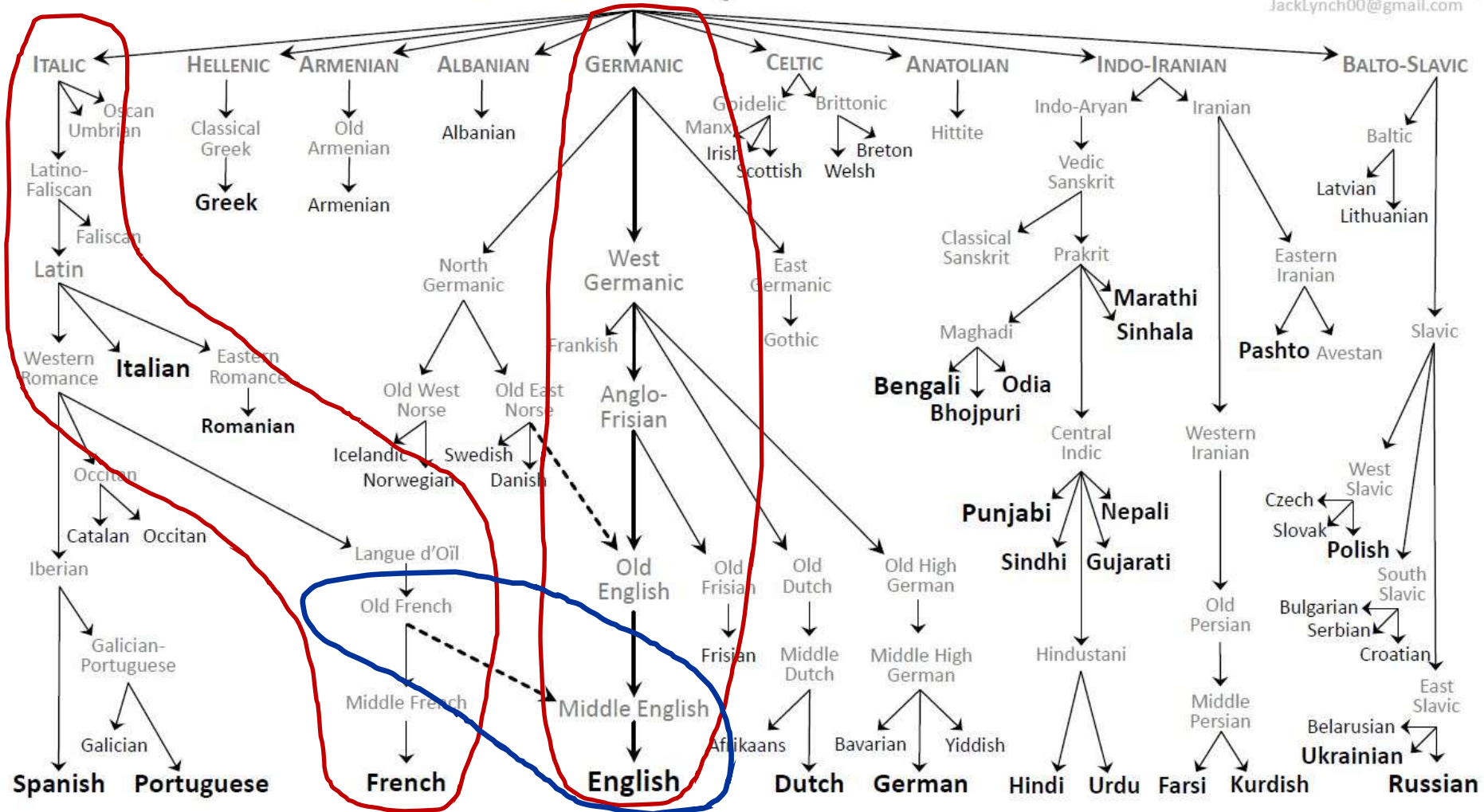
Constructed Language

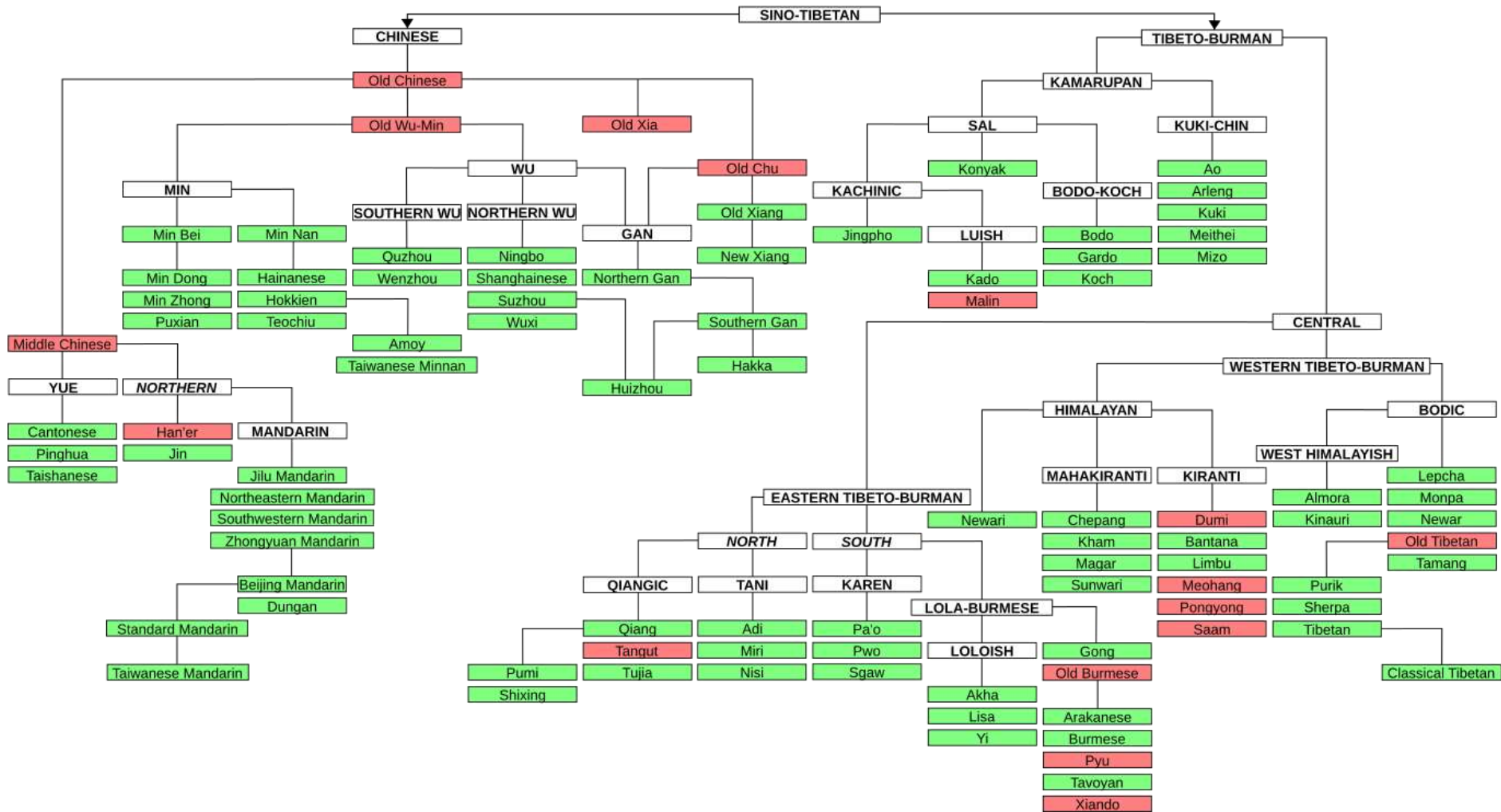
- created by individuals or groups.
- be precise and logical, and lacking the flexibility
- Grammar rules are strict, typically without exceptions or irregularities.

Natural Language Evolutions

- Language evolves continuously alongside the development of human society.
- Geographic isolation caused by tribal migrations leads to language evolving in different directions, eventually creating new languages.
- A language often absorbs elements (mainly vocabulary) from other geographically nearby languages, leading to convergent evolution.
- A **language family** is a group of languages related by descent from a common ancestral language, sharing similarities in phonology, vocabulary, and grammar.

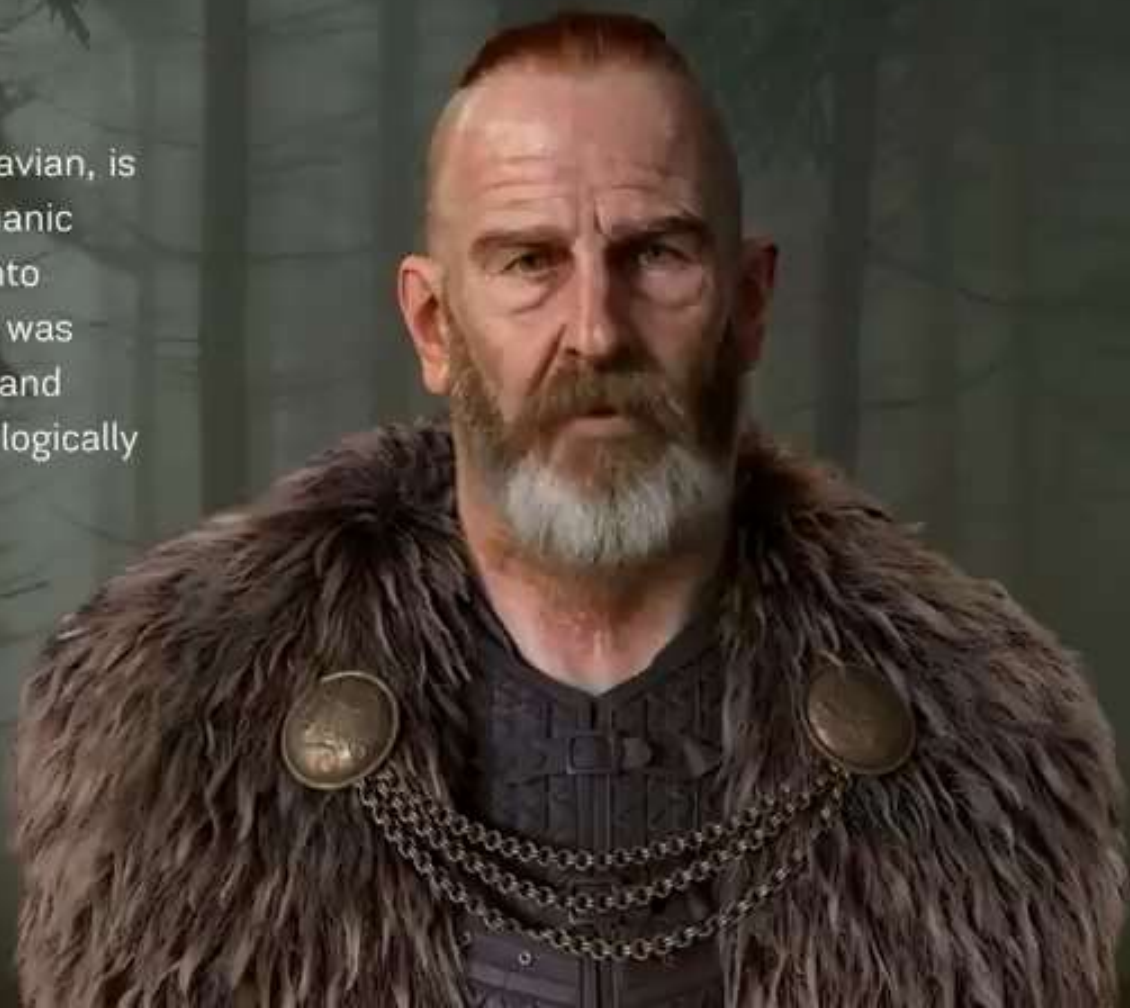






Old Norse

Old Norse, Old Nordic, or Old Scandinavian, is a stage of development of North Germanic dialects before their final divergence into separate Nordic languages. Old Norse was spoken by inhabitants of Scandinavia and their overseas settlements and chronologically coincides with the Viking Age.



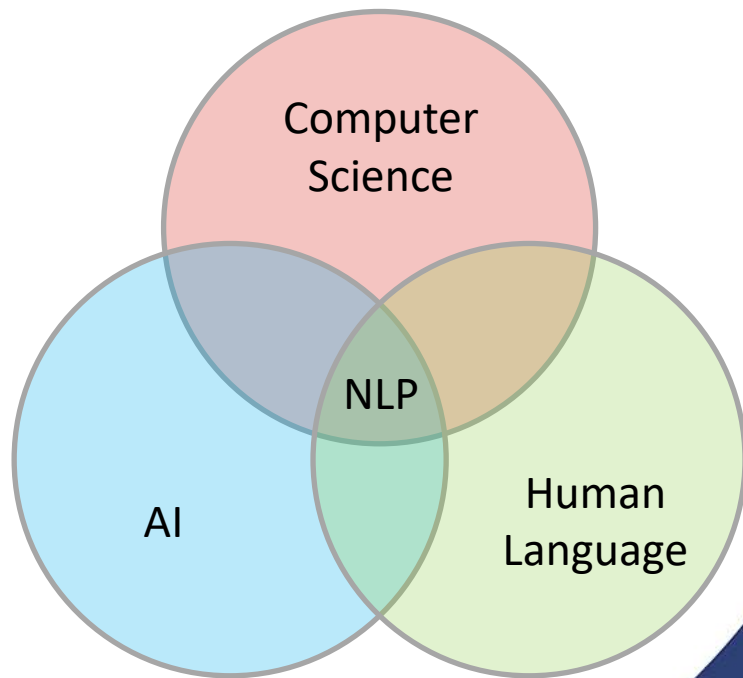
What is Natural Language Processing (NLP)

NLP

Natural Language Processing is a **branch of artificial intelligence** that deals with the interaction between computers and human language.

NLP encompasses the techniques and methods used to **understand**, **generate**, and **analyse** natural language **text** and **speech** data.

This course mainly introduces NLP knowledge from the perspective of Text. We only introduce speech when we specifically introduce speech-related topics.



NLP Tasks

- Segmentation
- Tokenisation
- Stemming
- Lemmatisation
- Part of Speech (POS) Tagging
- Named Identity Recognition (NER)
- ...

Segmentation

The process of dividing a sentence into its component sentences, usually along punctuation marks.

NLP is a machine learning technique that enables computers to interpret, process, and understand human language. It has lots of applications.



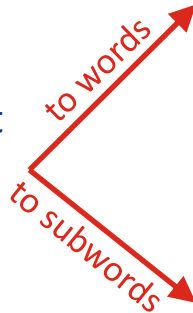
- NLP is a machine learning technique that enables computers to interpret, process, and understand human language.
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Tokenisation

The process of splitting sentences into their constituent **words** or **subwords**.

Splitting sentences into subwords also called **wordpiece tokenisation**.

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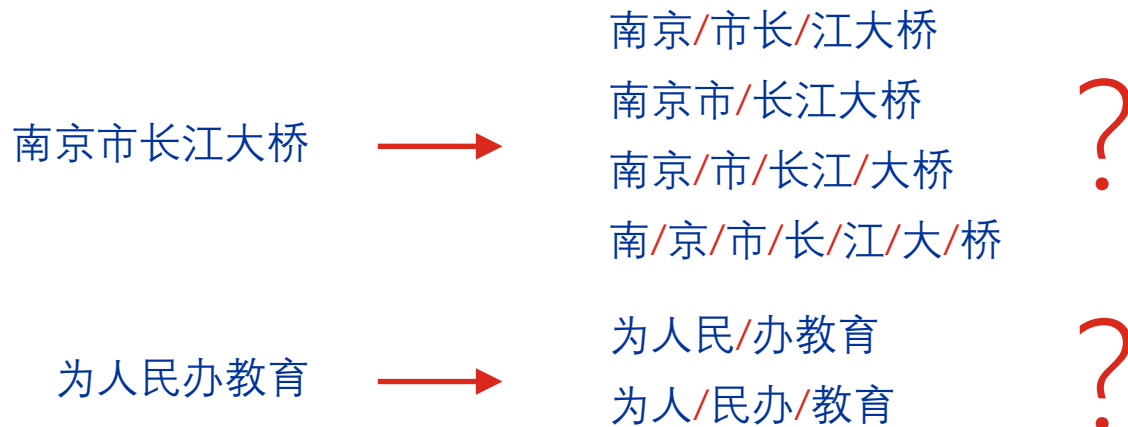
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Tokenisation for Chinese

Chinese word segmentation is often **more challenging**. Since there are no word boundaries in Chinese writing, the segmentation process needs to consider **granularity** and **ambiguity**.

New words frequently emerge in Chinese, making it a significant challenge to quickly identify them.



Stemming and Lemmatisation

Stemming involves removing affixes (such as prefixes and suffixes) from words to reduce them to their base form, known as the stem.

Lemmatisation, on the other hand, uses dictionaries and linguistic rules to reduce words to their base or dictionary form, known as the lemma.

stem		word		lemma
run	←	running	→	run
bett	←	better	→	good
cat	←	cats	→	cat
studi	←	studies	→	study

Part of Speech (POS) Tagging

It identifies which part of speech a word belongs to. It tags a word as a verb, noun, pronoun etc.

NLP is a machine learning technique that enables computers to interpret, process, and understand human language.



NLP/**NNP** is/**VBZ** a/**DT** machine/**NN**
learning/**NN** technique/**NN** that/**WDT**
enables/**VBZ** computers/**NNS** to/**TO**
interpret/**VB** ,/, process/**VB** ,/, and/**CC**
understand/**VB** human/**JJ** language/**NN** ./.

NNP (Proper Noun, singular)
VBZ (Verb, 3rd person singular present)
DT (Determiner)
NN (Noun, singular)
WDT (Wh-determiner)
NNS (Noun, plural)
TO (To)
VB (Verb, base form)
CC (Coordinating conjunction)
JJ (Adjective)

Named Entity Recognition (NER)

Classifying the words into subcategories, e.g. Person, Quantity, Location, Organisation ...

NLP is a machine learning technique that enables computers to interpret, process, and understand human language.



- NLP: (Entity: Technology)
- machine learning: (Entity: Technology)

NLP Applications



Chatbot



Speech
Recognition



Autocorrect



Machine
Translation

Course Arrangements

Teaching Arrangements

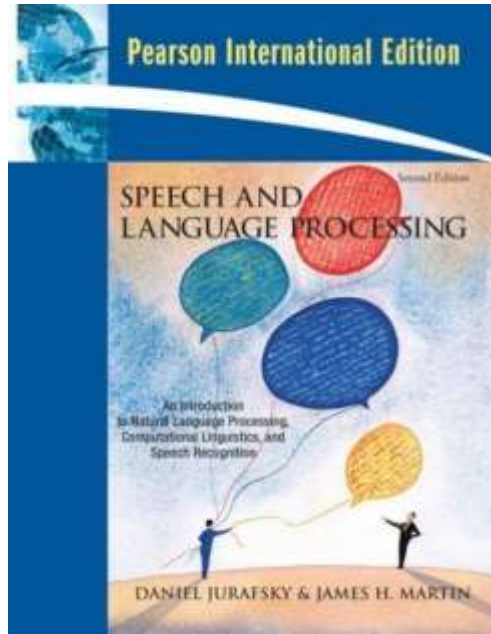
Teaching Arrangements:

- 32 theory sessions (across 10 weeks)
- 32 practical sessions (across 10 weeks)

Syllabus

- Formal linguistic models of English: word categories, sentence constituents, phrase-structure grammar rules, features. Modelling syntactic phenomena.
- Parsing: shift-reduce parsers, chart parsers, handling ambiguity, definite clause grammars.
- Semantics and pragmatics: meaning representations, reference, speech acts.
- Generation: Content determination, sentence planning, and realisation.
- Applications: grammar checking, machine translation, database interfaces, report generation, dictation.
- Speech: Hidden Markov Models, statistical language models, speech synthesis.

Textbook



Speech and Language Processing

Daniel Jurafsky
James H. Martin

<https://web.stanford.edu/~jurafsky/slp3/>

Theory Session Arrangements

- | | | | |
|----------------------------------|----------------------|-----------------------|--------------------|
| 1. Introduction | | 9. Neural Network LMs | } + Neural Network |
| 2. Linguistics Basics | } Linguistics | 10. Transformer | |
| 3. Context-free Grammars | | 11. BERT | } + Deep Learning |
| 4. N-gram LMs | } + Statistics | 12. GPT and LLM | |
| 5. Information Theory for NLP | | 13. Speech (ASR) | } Speech |
| 6. Naive Bayes & Classification | } + Machine Learning | 14. Speech (TTS) | |
| 7. Logistic Regression | | 15. Review1 | |
| 8. Word Semantics and Embeddings | | 16. Review2 | |

Practical Session Arrangements

Practical 1 & 2: Explain assessment requirements, build groups and Q&A

Practical 3: CFG + Parsing

Practical 4: N-gram language model

Practical 5: Information theory in NLP

Practical 6: Text classification and Naïve Bayes

Practical 7: Logistic Regression

Practical 8: Word embeddings

Practical 9: Neural Network

Practical 10: Transformer

Practical 11: BERT

Practical 12: GPT and LLMs

Practical 13 & 14: Speech

Practical 15: Review Q&A

Practical 16: Review Q&A

Assessment Methods

Assessment

- Final Exam (75%)
- Group programming assignment (25%)

Assessment

Group project: 4-6 people

Marking scheme: Group Report

Group formation deadline: 22:00 Friday, October 24, 2025

Report deadline: 22:00 Friday, December 5, 2025 (SCNU Local Time)

Ask Questions

We encourage students to ask questions at any time during working hours.

- **Just Ask:** feel free to interrupt me with your questions without the need to raise your hand, as we promote discussion in the classroom.
- You can also make use of the time during **lab sessions** or **breaks** to ask questions.
- Additionally, you are welcome to **send emails** with your queries, or pose them during the **Office Hour on Thursday afternoons** (it would be prudent to check the lecturer availability in advance due to the limited office space).

No Code Show Policy: We will **NOT** look at students' code as students are expected to know how to debug issues with their own code.

Conclusion

Natural Language vs. Artificial Language: Explored the complexity, flexibility, and ambiguity of natural languages compared to the precision of constructed languages.

NLP Overview: Introduced Natural Language Processing as the bridge between human language and AI, with key tasks like segmentation, tokenisation, and named entity recognition.

Course Structure

Assessment